

# AquaGuard — Bangladesh flood early-warning + IoT pond management

**What this is:** a final-year capstone project with two connected parts — (1) three machine-learning models that forecast flood risk for 30 river-gauge locations across Bangladesh using only free, public satellite/weather/river data, and (2) an ESP32-based IoT pond-management system (sensors + pumps) that this flood forecasting feeds into. This document is the starting point for anyone new to the project — a professor, a judge, a teammate, or future-you after time away — to get it running and understand what's here.

If you're going to present or demo this project, read this file first, then [Current status](#) so you know exactly what's real and what's still in progress — nothing here is oversold.

## Project map

| Folder / file              | What it is                                                                                                             |
|----------------------------|------------------------------------------------------------------------------------------------------------------------|
| manuals/                   | PDF versions of this file + the setup guides below, for offline/printed reading                                        |
| packages/                  | The 3 flood model APIs (FastAPI services) — see <a href="#">packages/SETUP_MANUAL.md</a> to run all three at once      |
| frontend-glass/            | The dashboard (glassmorphism look) — <b>recommended, most up to date version</b>                                       |
| frontend/                  | The same dashboard, neumorphic look — kept in sync with frontend-glass/ feature-for-feature                            |
| hardware/                  | The ESP32 + sensors + pumps rebuild — wiring guides, per-sensor test sketches, progress log                            |
| reports/                   | One rigorous PDF report per model (methodology, literature comparison, results, limitations)                           |
| backend/                   | The training pipeline that produced the models bundled inside packages/*/models/                                       |
| DECISIONS.md               | Every significant design decision made on this project, why, and what it costs — written for a pre-defense walkthrough |
| MODEL_BUILD_PLAN.md        | The full chronological build log — the detailed "how we got here," session by session                                  |
| hardware/HARDWARE_LOG.md   | Same idea as above, scoped to the physical hardware rebuild                                                            |
| hardware/FIREBASE_SETUP.md | Step-by-step guide for setting up the free cloud database that will link the hardware to the dashboard                 |
| PROJECT_FEATURE_IDEAS.md   | Whole-capstone feature ideas not yet committed to a build plan                                                         |

## Running the demo (zero prior context assumed)

### Step 1 — Start the 3 model services

```
cd packages
python -m venv .venv
```

Activate it (`.venv\Scripts\activate.bat` on Windows Command Prompt, `.venv\Scripts\Activate.ps1` on PowerShell, `source .venv/bin/activate` on Mac/Linux), then:

```
pip install -r requirements.txt
python run_all.py
```

Leave that terminal open. Full details, troubleshooting, and what each service does: [packages/SETUP\\_MANUAL.md](#).

## Step 2 — Open the dashboard

```
cd frontend-glass
python -m http.server 5502
```

Open <http://localhost:5502> in a browser. (Double-clicking `frontend-glass/index.html` directly usually works too — try that first if you'd rather not run a second terminal command; switch to the server method above only if "Use my location" doesn't prompt for permission.)

## Step 3 — Walk through it

1. Pick a station from the dropdown (or click "Use my location").
2. Click "> **Run full analysis**" — this calls all 3 model services at once: - **Flood risk classifier** — flood/no-flood risk at 24h/48h/72h ahead - **River discharge forecaster** — predicted river discharge (m<sup>3</sup>/s) at the same 3 horizons (its own prediction also feeds directly into the classifier's — see `DECISIONS.md §8` / `MODEL_BUILD_PLAN.md`'s "3-model pipeline" entry for how) - **Flood susceptibility** — how flood-prone the *terrain* is by nature, independent of current weather
3. Below all three, a **Pipeline summary** appears — 3 plain-language sentences stating what each model actually found (deliberately *not* a single LOW/HIGH verdict — see `MODEL_BUILD_PLAN.md`'s entry on why that design was chosen, so the reader weighs the evidence themselves).

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## Current status

Written plainly, not to oversell what's finished:

- [done] **All 3 flood models**: trained, tested, documented, each with its own PDF report in `reports/`.
- [done] **Dashboard**: real, live calls to all 3 models; the 3-model pipeline (cascade + combined summary) is built and working.
- [in progress] **Hardware sensor readings on the dashboard**: currently **demo data** (a random walk, clearly labeled as such in the UI) — 4 of 7 physical sensors are individually wired and confirmed working (see `hardware/HARDWARE_LOG.md`), but not yet streaming live to the dashboard.
- [in progress] **Pump control on the dashboard**: currently **simulated** (changes on-screen state only, no real relay control) — the relay hardware itself is sketch-tested but not yet wired to real pumps.
- [in progress] **Live ESP32 → dashboard link + pump automation logic**: in progress. `hardware/FIREBASE_SETUP.md` is the first step (setting up the free database that will carry live sensor data and pump commands between the physical device and the dashboard).
- [in progress] **pH calibration page** (`ph-calibration.html`, linked from the dashboard header): code and firmware side (`hardware/rebuild/07_full_reintegration/AquaGuard_v2.ino`) are written, but need a real Firebase project (see above) and a real ESP32 flash before this actually works end to end.
- [in progress] **Public hosting (Vercel)**: not yet deployed — the dashboard currently runs locally only (Steps 1-2 above). The 3 model services are also local-only by design for now (a deliberate, documented choice — see

## For a technical deep-dive / defense

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- **reports/** — one PDF per model (`flood-risk-classifier/`, `discharge-forecaster/`, `flood-susceptibility/`), each covering literature review, methodology, results vs. published benchmarks, limitations, and reproducibility steps. `reports/legacy/` holds the original combined pre-split report, kept for history.
- **DECISIONS.md** — every consequential decision (data sources, label design, model family, station coverage, etc.), why it was made, what alternatives were considered, and what limitation it leaves — meant to be read start to finish by someone who wasn't in the room.
- **MODEL\_BUILD\_PLAN.md** — the raw, dated, chronological working log of the whole build, for anyone who wants the full "how we actually got here" including dead ends and negative results.

